

Total no. of Pages:

Roll no.

Degree: B. Tech (CSIOT & CSDA) Semester: II

END-SEMESTER EXAMINATION, APRIL-MAY 2025

Course Code: CIEPC203 / CBEP203

Course Title: DIGITAL LOGIC DESIGN

Max. Marks: 50

Duration: 03 Hours

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following	5	1
1a	State D-morgan's laws and verify using truth table.	3 + 2	
1b	Given numbers: $x = 164$ and $y = 75$; (1) find binary and hexadecimal value of x and y (2) evaluate $(x - y)$ using 2's complement.	5	
1c	Given numbers: $p = 265$ and $q = 973$; convert them to BCD and find their BCD sum.		
Q 2	Attempt any 2 parts of the following	5	
2a	Minimize the Boolean function using K-map and implement using only NAND gates:- $F(A, B, C, D) = \sum \min(1, 3, 6, 11, 12, 14) + d(0, 4, 9, 15)$.	5	
2b	Design a BCD adder using two 4-bit parallel adders.	5	
2c	Using an 8:1 MUX, implement the function: $F(P, Q, R, S) = \sum \min(0, 1, 4, 7, 8, 11, 12, 14)$		
Q 3	Attempt any 2 parts of the following	5	
3a	Draw and explain 4-bit PISO shift register, using JK Flip-flops	5	
3b	Design a 3-bit synchronous down counter using D Flip-flops	5	
3c	Design a logic circuit to detect the presence of the 4-bit sequence: 0 1 1 0		
Q 4	Attempt any 2 parts of the following	2.5 + 2.5	
4a	Write a VHDL module code for: (a) JK-F/F and (b) D-F/F	5	
4b	Write a VHDL module code 4-bit binary counter	5	
4c	Write a VHDL module code for 8:1 Multiplexer		
Q 5	Attempt any 2 parts of the following	5	
5a	Explain the following parameters of logic families: (i) Propagation delay, (ii) Noise margin, (iii) fan-in and fan-out	5	
5b	Draw and explain CMOS transistor level schematic of 3-inputs NOR gate	5	
5c	Write a short note on PLA		